

# Information Retrieval Lecture Notes: TF-IDF & Vector Space Model

DEPARTMENT OF INFORMATION TECHNOLOGY - COURSE: INFORMATION RETRIEVAL

Unit 2: Text Preprocessing, Term Weighting, and Vector Space Retrieval

## 1. Text Preprocessing in Information Retrieval:

Raw documents undergo tokenization, case normalization, punctuation removal, and stop-word filtering using curated stop-word dictionaries.

## 2. Inverted Index Data Structure:

An inverted index is an essential IR data structure mapping each dictionary term to a postings list containing document identifiers and term frequencies.

## 3. Term Frequency - Inverse Document Frequency (TF-IDF):

TF-IDF measures the importance of a term within a document relative to the entire corpus. Term Frequency (TF) captures local importance, while Inverse Document Frequency (IDF) penalizes common terms across all documents:  
 $IDF(t) = \log(N / DF(t))$ .

## 4. Vector Space Model & Cosine Similarity:

Documents and queries are represented as high-dimensional TF-IDF vectors. Retrieval ranking evaluates the cosine similarity between the query vector and candidate document vectors.